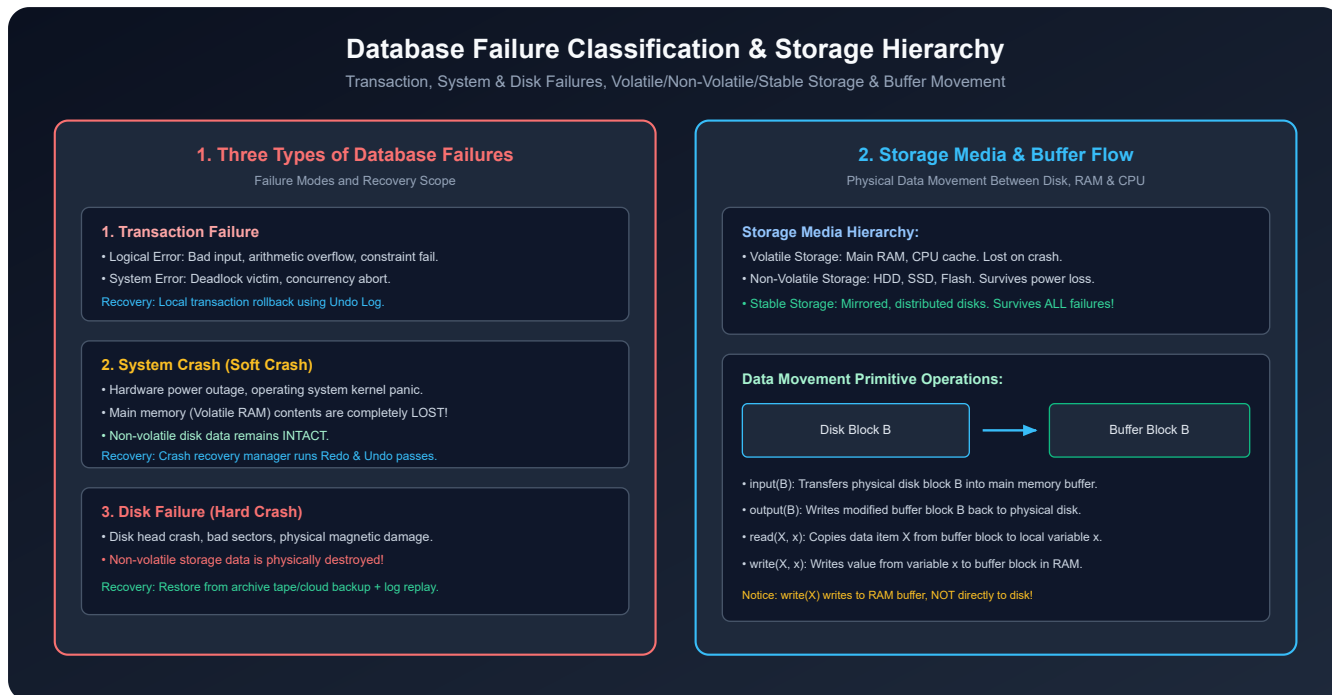


Module 09: Failure Types & Storage Structures



> **Course:** Database Management Systems (AKTU Code: **BCS501**)

****Topic Focus:**** Failure Classification (Transaction, System Crash, Disk Failure), Storage Media Hierarchy (Volatile, Non-Volatile, Stable Storage), Buffer Management and Primitive Data Movement Operations (`input`, `output`, `read`, `write`).

1. Classification of Database Failures

Database system me failures ko unke scope aur loss ke according 3 categories me divide kiya jaata hai:

1.1 Transaction Failure

Logical Error: Transaction business logic ya application error ki wajah se aage nahi badh sakti (e.g., negative balance check, divide-by-zero, foreign key violation).

System Error: Concurrency control manager deadlock break karne ke liye transaction ko victim select karke abort kar deta hai.

Recovery: Local transaction rollback through undo log.

1.2 System Crash (Soft Failure)

Cause: Power failure, OS kernel panic, hardware motherboard crash.

Consequence: Main memory (RAM) ka poora volatile content destroy ho jaata hai. Disk par store data safe rehta hai.

Assumption (Fail-Stop Model): Hardware crash hone par database hardware turant halt ho jaata hai bina disk data ko corrupt kiye.

1.3 Disk Failure (Hard Failure)

Cause: Disk read/write head crash, magnetic degradation, bad disk blocks.

Consequence: Non-volatile storage permanently damage ho jaata hai.

Recovery: Periodic archival database backups (Dump) aur stable log copies se full restore.

2. Storage Media Hierarchy

1. Volatile Storage:

- RAM, cache memory. Extremely fast ($O(1)$ ns access). Lost during power cuts.

2. Non-Volatile Storage:

- Magnetic hard disks, solid-state drives (SSD), flash memory. Retains data across crashes.

3. Stable Storage:

- Theoretical ideal storage jo **kisi bhi failure me kabhi corrupt nahi hota**.
- Physically implemented using **RAID 1 / Mirroring** across independent disk controllers aur geographically remote backup sites.

3. Data Movement and Buffer Management Architecture

Database disk par block size me store hota hai (e.g., 4KB blocks). Transaction CPU registers me execute hoti hai. Inke beech data ka flow 4 fundamental primitive operations se hota hai:

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[Physical Disk]

▲ |

output | | input

| ▼

[Main Memory Buffer]

▲ |

write | | read

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[Transaction Local Registers]

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input(B): Disk block B ko read karke memory buffer block me laata hai.

read(X, x): Buffer block se item X ko transaction ke local variable x me copy karta hai.

write(X, x): Local variable x ki value ko RAM ke buffer block me likhta hai. *(Caution: Yeh abhi disk par nahi gaya hai!)*

output(B): RAM ke dirty buffer block B ko disk par permanently flush karta hai.

4. Architectural Diagram
